

Deep Networks

Bias, Variance and Double Descent

Introduction to Deep Learning · Class 17

Outline

What We Are Measuring

The Decomposition

The Classical Trade-off

Double Descent

What This Changes

What We Are Measuring

From training error to test error

Classes 13 to 15 asked how to reach a minimum of $E(\mathbf{w})$ on the training set. Nothing so far has asked what that buys on data we have not seen.

Settled in Classes 13–15

how to descend $E(\mathbf{w})$ efficiently

how conditioning governs the rate

how to make the landscape kind

Open today

what the error on unseen data is made of

how that error moves with model size

why very large models are not ruined

Today is diagnostic. We decompose the test error exactly, then find a measurement the decomposition does not predict.

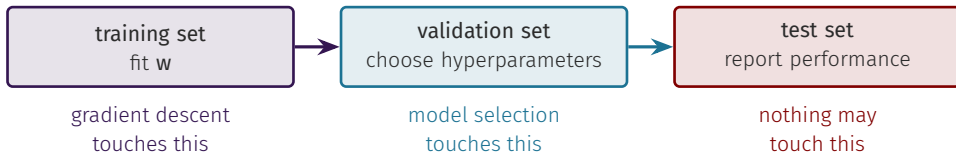
Notation for today

$\mathcal{D} = \{\mathbf{x}_n, Y_n\}$	one training set of size N	$\mathbb{E}_{\mathcal{D}}[\cdot]$	average over training sets
$\hat{y}(\mathbf{x}; \mathcal{D})$	the model fitted to that set	$\bar{y}(\mathbf{x})$	$\mathbb{E}_{\mathcal{D}}[\hat{y}(\mathbf{x}; \mathcal{D})]$, the ensemble mean
$h(\mathbf{x}) = \mathbb{E}[Y \mathbf{x}]$	the regression function	p	number of learnable parameters
σ^2	variance of Y about $h(\mathbf{x})$	$\gamma = p/N$	parameters per observation

- Bishop writes $h(\mathbf{x})$ and Prince writes $\mu[\mathbf{x}]$ for the same object; both mean the conditional mean of the target

Bishop & Bishop (2024), Eq. 4.41; Prince (2023), Eq. 8.1.

Three data sets, three different objectives



- Hyperparameters chosen on the test set would report a number that cannot be reproduced on new data
- The hyperparameter space is discrete and conditional, so gradient descent cannot search it

Prince (2023), §8.5.

The Decomposition

The best predictor there is

Proposition 1 (The conditional mean minimises squared loss)

For squared loss the expected loss of any function f splits as

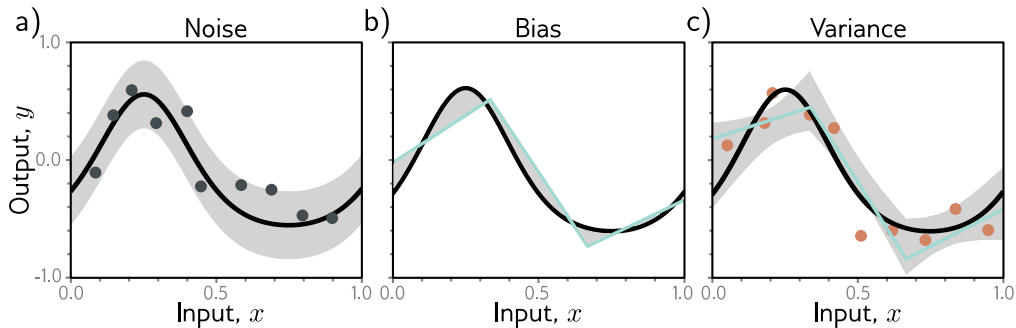
$$\mathbb{E}[L] = \int \{f(\mathbf{x}) - h(\mathbf{x})\}^2 p(\mathbf{x}) d\mathbf{x} + \int \text{var}[Y | \mathbf{x}] p(\mathbf{x}) d\mathbf{x},$$

so it is minimised by $f = h$, and the second term does not depend on f at all.

- The second term is the **noise**: the price of predicting a random variable with a function
- No model, no data set and no algorithm can reduce it
- Everything today concerns the first term

Bishop & Bishop (2024), Eqs. 4.39, 4.42.

Where the error comes from



(a) noise: the targets scatter about the true function, leaving a floor no model can cross. (b) bias: the best available fit still departs from it. (c) variance: a fit to a finite sample departs again, differently each time.

Prince (2023), Fig. 8.5.

The decomposition

Theorem 2 (Bias–variance–noise)

For squared loss, at each $\mathbf{x}$,

$$\underbrace{\mathbb{E}_{\mathcal{D}} \mathbb{E}_Y [(\hat{y}(\mathbf{x}; \mathcal{D}) - Y)^2]}_{\text{expected test error}} = \underbrace{\mathbb{E}_{\mathcal{D}} [(\hat{y}(\mathbf{x}; \mathcal{D}) - \bar{y}(\mathbf{x}))^2]}_{\text{variance}} + \underbrace{(\bar{y}(\mathbf{x}) - h(\mathbf{x}))^2}_{\text{bias}^2} + \underbrace{\sigma^2}_{\text{noise}}.$$

- **Bias** is how far the *average* model is from the truth — a property of the model class
- **Variance** is how far a *single* model is from that average — a property of the sample
- The three terms are additive here; for other losses they interact

Prince (2023), Eq. 8.7; Bishop & Bishop (2024), Eqs. 4.45–4.49.

Proof, in two identical steps

- **Add and subtract $h(\mathbf{x})$ inside the square:** $(\hat{y} - Y)^2 = \{(\hat{y} - h) + (h - Y)\}^2$.
- Expanding gives a cross term $2(\hat{y} - h)(h - Y)$, whose expectation over Y vanishes because $\mathbb{E}_Y[Y] = h(\mathbf{x})$ by definition.
- What survives is $\mathbb{E}_Y[(\hat{y} - Y)^2] = (\hat{y} - h)^2 + \sigma^2$. The noise has separated.
- Now repeat the same move on $(\hat{y} - h)^2$, this time adding and subtracting $\bar{y}(\mathbf{x})$ and averaging over $\mathcal{D}$.
- The cross term $2(\hat{y} - \bar{y})(\bar{y} - h)$ vanishes for the same reason: $\mathbb{E}_{\mathcal{D}}[\hat{y}] = \bar{y}$ by definition.
- This leaves variance + bias², and with the σ^2 from step 3 the theorem follows. □

One algebraic move, used twice: add and subtract the mean, and the cross term dies because a deviation from a mean has zero mean.

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What the decomposition is and is not

Corollary 3 (The floor)

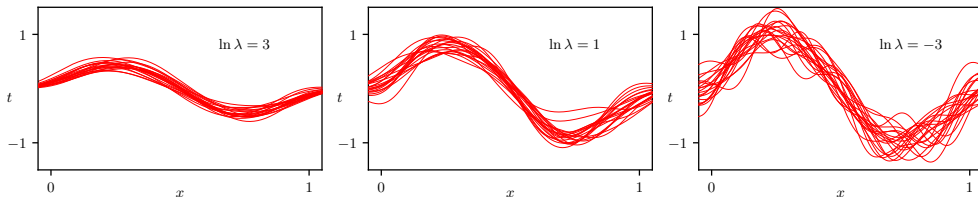
Since variance ≥ 0 and bias² ≥ 0 , the expected test error is at least σ^2 , with equality only if the fitted model equals h for every training set.

- Both terms are averages over an **ensemble** of training sets; in practice we hold exactly one
- So neither term is measurable from the data we have, which is why validation exists
- Given several independent training sets one would combine them rather than average over them

Bishop & Bishop (2024), §4.3; Prince (2023), §8.2.2.

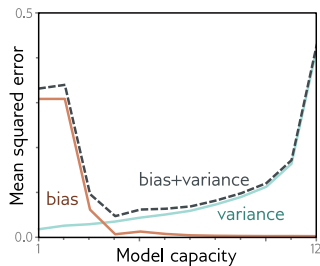
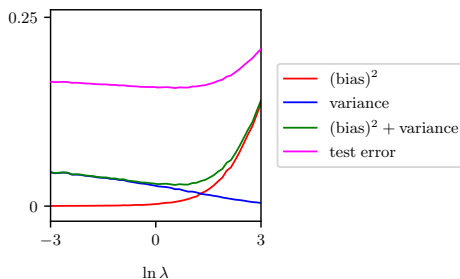
The Classical Trade-off

Two ways to spend capacity



Twenty of one hundred fits, each to its own sample of $N = 25$ points, at three values of the penalty: $\ln \lambda = 3$, 1 and -3 . The spread across the red curves is the variance. Bishop & Bishop (2024), Fig. 4.7.

The trade-off, measured



The same statement in both books. Left: against the penalty $\ln \lambda$, with the test error overlaid; the minimum of $\text{bias}^2 + \text{variance}$ sits where the test error is smallest. Bishop & Bishop (2024), Fig. 4.8. Right: against model capacity. Prince (2023), Fig. 8.9.

What reduces each term

Term	Reduced by	At the cost of
noise σ^2	nothing	—
bias ²	more capacity	more variance
variance	more data	data is what is scarce
variance	a penalty on $\mathbf{w}$	more bias

- More data lowers variance without raising bias, which is why it is the one intervention with no trade
- Every other row is a trade, and the classical conclusion is that some interior capacity is best

The classical prescription

The received view

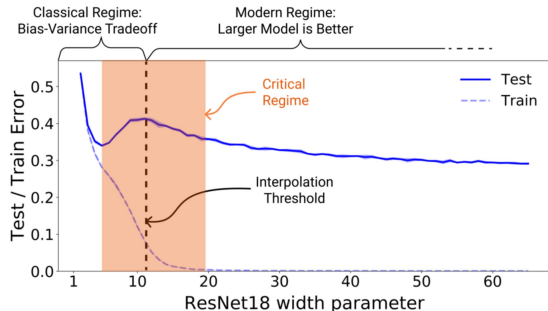
Test error is U-shaped in model complexity. Choose the model whose capacity matches the size of the data set: too small and bias dominates, too large and variance dominates. Very large models trained on modest data sets are therefore expected to generalize badly.

- A correct reading of the decomposition, and of every curve on the previous slide
- It is also the standard advice in classical statistics
- Modern networks have far more parameters than training points

Bishop & Bishop (2024), §9.3.2.

Double Descent

What is actually measured



ResNet18 on an image classification task; the horizontal axis is a width parameter governing the number of weights. Training error falls monotonically; test error falls, rises, and falls again, continuing to fall after the training error has reached zero. Bishop & Bishop (2024), Fig. 9.9, from Nakkiran et al. (2019).

Reading the curve

- **Axes.** Capacity is a width parameter setting the number of parameters; the training set is fixed. Dashed is training error, solid is error on unseen data, and only the solid one matters
- **Threshold.** The model can fit every training point exactly; the training error reaches zero and stays there
- **Classical regime, left of the band.** Test error falls as bias falls, then rises as variance rises: the U-curve, with its **classical minimum**
- **Critical regime, the band.** Exactly one interpolant exists and it is enormous; the test error peaks
- **Modern regime, right of the band.** Many interpolants exist and the smallest is chosen. Test error descends again to the **modern minimum** — below the classical one, although the training error has been zero since the threshold

The interpolation threshold

Definition 4 (Interpolation threshold)

The capacity at which the model first attains zero training error. Below it the data constrain the fit; above it a family of exact fits exists and the algorithm chooses among them.

Definition 5 (Effective model complexity)

The largest training set size on which a given training procedure reaches approximately zero training error. Double descent appears where it crosses N .

- Defined for the **procedure**, not the architecture: the optimiser and the epoch count are part of it

Belkin et al. (2019); Nakkiran et al. (2019).

The risk, in closed form

Theorem 6 (Ridgeless least squares, isotropic design)

Let $Y = X\mathbf{w} + \varepsilon$ with rows of X drawn from $\mathcal{N}(\mathbf{0}, \mathbf{I}_p)$ and $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, and let $\hat{\mathbf{w}}$ be the minimum-norm least-squares solution. As $N, p \rightarrow \infty$ with $p/N \rightarrow \gamma$, the excess risk tends to

$$R(\gamma) = \begin{cases} \frac{\sigma^2 \gamma}{1 - \gamma}, & \gamma < 1, \\ \|\mathbf{w}\|^2 \left(1 - \frac{1}{\gamma}\right) + \frac{\sigma^2}{\gamma - 1}, & \gamma > 1. \end{cases}$$

- Both branches diverge as $\gamma \rightarrow 1$: the peak is a pole, not an accident of any particular experiment

Hastie, Montanari, Rosset & Tibshirani (2022), Thm. 1.

Reading the two branches

- Below the threshold there is one least-squares solution and it must absorb the noise in $N - p$ fewer degrees of freedom than it has observations; the risk $\sigma^2\gamma/(1 - \gamma)$ grows as that slack disappears.
- At $\gamma = 1$ the design is square and almost singular. Exactly one vector interpolates, and it is forced to be enormous.
- Above the threshold the interpolating set is an affine subspace of dimension $p - N$, and minimum norm picks one point of it.
- Growing γ enlarges that subspace, so the chosen point can be smaller: the noise term $\sigma^2/(\gamma - 1)$ falls.
- The price is the bias term $\|\mathbf{w}\|^2(1 - 1/\gamma)$, which rises to $\|\mathbf{w}\|^2$. The second descent is the race between the two.

Two solutions fit the data equally well; the risk depends entirely on which of them the algorithm returns.

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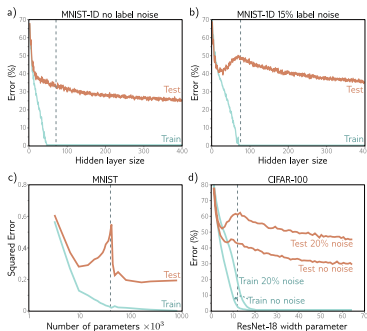
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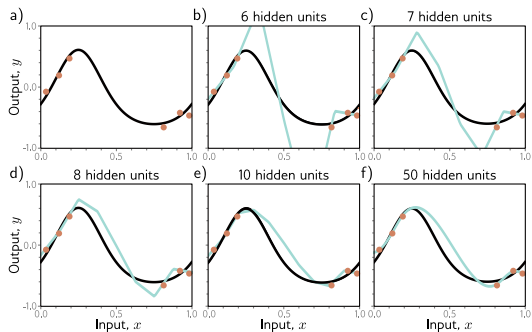
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The same shape across four data sets



- (a) MNIST-1D without label noise: no peak. (b) the same with 15% of labels randomised: the peak appears.
(c) MNIST with a shallow network. (d) CIFAR-100 with ResNet18. How pronounced the peak is depends on the noise in the labels. Prince (2023), Fig. 8.10.

Smoothness is what extra capacity buys



The smoothest function that still passes through every training point, as hidden units are added. With just enough capacity the curve must contort itself; with more, it can interpolate smoothly across the gap. Prince (2023), Fig. 8.11.

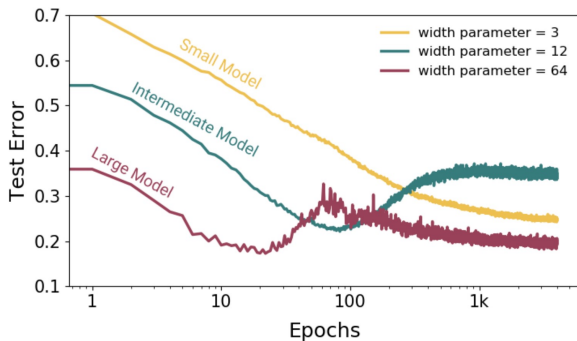
Why sparse data makes this matter

- Past the threshold the data are fitted exactly, so extra capacity can only change the model **between** the training points
- In high dimensions almost everything is between them: quantising 40 inputs into 10 bins gives 10^{40} cells for 10^5 examples
- So what decides the test error is behaviour no training point constrains
- A procedure's tendency to prefer one such completion is its **inductive bias**; here, a preference for small norm and so for smoothness

What produces that preference, and how to install one deliberately, is the subject of Class 18.

Prince (2023), §8.4.1.

Capacity is not the parameter count



Test error against training epochs for three model sizes. Effective complexity grows with the number of epochs, so the large model traverses the same three regimes in time that the previous figure traversed in width. Bishop & Bishop (2024), Fig. 9.10, from Nakkiran et al. (2019).

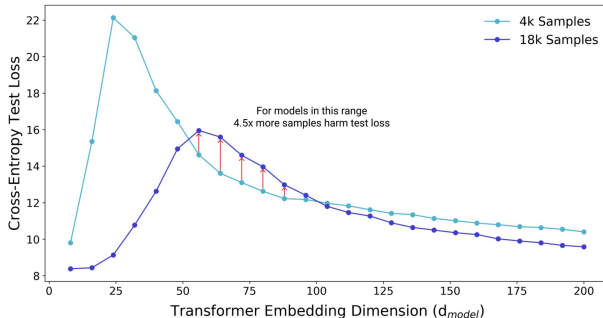
The same curve along three different axes

Axis varied	Peak sits at	Name
number of parameters, N fixed	$p = N$	model-wise
training epochs, p and N fixed	$d_{\text{eff}} \approx N$	epoch-wise
training set size, p fixed	$N = p$	sample-wise
inverse penalty $1/\lambda$	where the fit first interpolates	—

- All four are the same statement about effective complexity crossing the amount of data
- Epoch-wise is why a large model can look worse after longer training and better again after longer still

Nakkiran et al. (2019); Yilmaz & Heckel (2022).

More data can make a model worse



Test loss for a transformer against embedding dimension, at two training set sizes. The larger set is better overall, but over the range marked by the arrows 4.5 $\times$ more data gives a *worse* model. Bishop & Bishop (2024), Fig. 9.11, from Nakkiran et al. (2019).

What This Changes

What survives and what does not

Claim	Status
$\text{error} = \text{bias}^2 + \text{variance} + \text{noise}$	exact, and unaffected
noise is irreducible	exact, and unaffected
bias falls with capacity	holds
variance rises with capacity	holds <i>below</i> the threshold only
test error is U-shaped in capacity	false past the threshold
a model with $p > N$ must over-fit	false
more data cannot hurt	false near $N = p$

The decomposition is a theorem and stays true. What fails is the empirical claim about how variance behaves once the model can interpolate.

Choosing capacity when the curve has two minima

- Neither term is measurable from one training set, so capacity is chosen by **validation**, not by theory
- In the classical regime the validation curve has an interior minimum, and the search is a line search
- In the modern regime it falls with capacity as far as one can afford, and the binding constraint becomes compute
- The regime to avoid is the critical one, $p \approx N$

Either stay well below the threshold or go well past it. Sitting on it is the one choice the measurements rule out.

Prince (2023), §8.5; Nakkiran et al. (2019).

Takeaways

1. For squared loss the expected test error is exactly $\text{bias}^2 + \text{variance} + \text{noise}$, and the proof is one algebraic move used twice (Theorem 2)
2. Noise is a floor no procedure can cross (Corollary 3)
3. Below the interpolation threshold the classical U-curve is real and measurable
4. At the threshold there is exactly one interpolant, its norm diverges, and so does the risk — exactly, by Theorem 6
5. Past the threshold there is a subspace of interpolants; minimum norm picks a smooth one and the risk descends again
6. Model-wise, epoch-wise and sample-wise double descent are one phenomenon: effective complexity crossing N
7. Capacity is a property of the procedure, and it grows while you train

Class 18 asks what we can add to the objective, and what the algorithm is already adding, to control that choice deliberately.

References i

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Questions?